CHEM 1301 Driller 9-4 — Answer Key

Worked solutions with course-material citations

Every numeric answer was recomputed by `_scripts/build_driller_9_4.py`, which builds this key and the driller from the same data. Atomic masses are periodic-table values (H 1.008, C 12.01, N 14.01, O 16.00, …). Citations name the slide (same number in the `.pdf` and `.pptx`) where the content is taught, plus the closest homework screenshot.

How to read the solutions. Each calculation is a chain of conversion factors written want-over-have. The label under each factor says where its numbers come from (a molar mass, or two coefficients from the balanced equation), and the crossed-out units are the ones that cancel. Study guide section 9.0 explains this format.

Score bands. 14–15 solid on 8.7 and Chapter 9 · 11–13 rework the study guide sections cited for the questions you missed · 10 or below go back through study guide sections 8.7 and 9.2–9.7 before the next driller.

Q	Ans	Answer	Type	Topic (source)
1	**b**	Multiply every element's result by 3, giving $C_3H_4O_3$	concept	A result that isn't a whole number (Ch 8 sl. 43)
2	**d**	CH_2O	calc	Molecular to empirical formula (Ch 8 sl. 39–40)
3	**e**	$C_9H_8O_4$	calc	Percent composition to empirical formula (×4) (Ch 8 sl. 42–43)
4	**a**	$K_2Cr_2O_7$	calc	Percent composition to empirical formula (×2) (Ch 8 sl. 35–38, 42–43)
5	**d**	Na_2SO_3	calc	Grams to empirical formula (Ch 8 sl. 45–46)
6	**e**	C_9H_{12}	calc	Empirical to molecular formula (Ch 8 sl. 47–49)
7	**c**	0.600 mol	calc	From percent composition to a mole ratio (Ch 8 sl. 42–49; Ch 9 sl. 9–11)
8	**a**	The coefficients of A and B in the balanced equation	concept	What a mole–mole conversion needs (Ch 9 sl. 5–7, 10)
9	**a**	1.20 mol	calc	Mole–mole conversion (Ch 9 sl. 10–11)
10	**c**	0.399 mol	calc	Grams to moles of another substance (Ch 9 sl. 14, 18)

Q	Ans	Answer	Type	Topic (source)
11	**e**	17.0 g	calc	Grams to grams (Ch 9 sl. 14, 19–20)
12	**d**	9.42 g	calc	Balance first, then grams to grams (Ch 9 sl. 14, 16, 19)
13	**b**	O_2 is limiting, and 9.40 g of ZnS is left over	calc	Limiting reactant and leftover (Ch 9 sl. 21–24)
14	**b**	15.9 g	calc	Theoretical yield from two masses (Ch 9 sl. 24–27)
15	**c**	88.1%	calc	Percent yield (Ch 9 sl. 28–32)

Mix: 2 concept / 13 calculation · Ch 8: 6 · Ch 9: 9

1. A result that isn't a whole number — b) Multiply every element's result by 3, giving $C_3H_4O_3$

Source: Chapter 8 post, slide 43 (step 4: multiply by the smallest integer that gives whole numbers), slides 42, 46
Homework match: HW21 — hw21ex2.png (empirical formula from percent composition)
What it tests: Step 4 of the empirical-formula method: .33 means thirds, multiply **every** element by 3.

Step 1: is 1.33 a whole number? "Whole" means within about 0.1 of an integer. 1.33 is 0.33 away from 1, so it is **not** whole, and you must not round it.

Step 2: which multiplier? A decimal near .33 (or .67) is a third, so multiply by **3**: $1.33 \times 3 = 3.99 \approx 4$.

Step 3: apply it to every element. C $1.00 \times 3 = 3$, H $1.33 \times 3 = 4$, O $1.00 \times 3 = 3$, giving $\mathbf{C_3H_4O_3}$. This is the vitamin C example in study guide 8.7 (Type B, example C).

Why the other choices are wrong:

- a) Divide every result by 1.33 again: That would make C and O fractions (0.75). Step 4 multiplies; it never divides again.
- c) Multiply every result by 2, giving $C_2H_3O_2$: $1.33 \times 2 = 2.67$, still not whole. A .33 means thirds, so multiply by 3.
- d) Multiply only the H result by 3, giving CH_4O: That changes the ratio. The same multiplier goes on **every** element.
- e) Round H to 1, giving CHO: Only results within about 0.1 of a whole number may be rounded. 1.33 is 0.33 away.

2. Molecular to empirical formula — d) CH_2O

Source: Chapter 8 post, slide 39 (empirical formula = smallest whole-number ratio), slide 40
Homework match: HW21 — hw21ex3.png (relating empirical and molecular formulas)
What it tests: Type A (study guide 8.7): use the **largest** common divisor, not just 2.

Step 1: list the subscripts. C 4, H 8, O 4.

Step 2: find the largest number that divides all three evenly. $4 \div 4 = 1$, $8 \div 4 = 2$, $4 \div 4 = 1$. Nothing larger than 4 divides 4, so the divisor is **4**. (Dividing by 2 gives $C_2H_4O_2$, which can still be reduced, so it isn't the smallest ratio.)

Step 3: divide every subscript by 4. CH_2O.

Why the other choices are wrong:

- a) CH_4O: Divided C and O by 4 but H only by 2.
- b) $C_4H_8O_4$: That is the molecular formula; 4, 8, and 4 share a factor of 4.
- c) CHO: Divided H by 8 instead of 4. Every subscript is divided by the **same** number.
- e) $C_2H_4O_2$: Divided by 2, but 2, 4, and 2 can still all be divided by 2. Use the **largest** common divisor, 4.

3. Percent composition to empirical formula (×4) — e) $C_9H_8O_4$

Source: Chapter 8 post, slides 42–43 (Steps for Determining the Empirical Formula; step 4: multiply when not whole)
Homework match: HW21 — hw21ex2.png (empirical formula from percent composition)
What it tests: The 4-step method: grams → moles → divide by the smallest → whole numbers (multiply every element if needed).

Step 1: percent to grams. Pretend you have exactly **100 g** of the compound, so each percent becomes that many grams: 60.00 g C, 4.48 g H, 35.52 g O.

Step 2: grams to moles for each element. Divide by each element's own atomic mass (periodic table):

$$60.00 \text{ g } \cancel{\text{C}} \times \underbrace{\frac{1 \text{ mol C}}{12.01 \text{ g } \cancel{\text{C}}}}_{\text{atomic mass of C}} = 4.996 \text{ mol C}$$

$$4.48 \text{ g } \cancel{\text{H}} \times \underbrace{\frac{1 \text{ mol H}}{1.008 \text{ g } \cancel{\text{H}}}}_{\text{atomic mass of H}} = 4.444 \text{ mol H}$$

$$35.52 \text{ g } \cancel{\text{O}} \times \underbrace{\frac{1 \text{ mol O}}{16.00 \text{ g } \cancel{\text{O}}}}_{\text{atomic mass of O}} = 2.220 \text{ mol O}$$

Step 3: divide every mole value by the smallest one. The smallest mole value is 2.220 mol (O). Divide **each** element's moles by it:

Element	Moles (step 2)	÷ smallest (2.220 mol O)	× 4	Subscript
C	4.996	2.25	9.00	9
H	4.444	2.00	8.01	8
O	2.220	1.00	4.00	4

Step 4: whole numbers? 2.25 is not within 0.1 of a whole number. It ends in a decimal near .25 or .75 (quarters), so multiply **every** result by 4, including the elements that were already whole (last columns of the table). Empirical formula: $\mathbf{C_9H_8O_4}$.

Why the other choices are wrong:

- a) C_2H_2O: Rounded the fractional result instead of multiplying (only results within 0.1 of a whole number may be rounded).
- b) $C_{18}H_{16}O_8$: Not the **smallest** whole-number ratio; every subscript can still be divided by 2.
- c) C_9H_2O: Multiplied only the fractional element by 4. The same multiplier goes on **every** element.
- d) $C_{13}HO_8$: Used the masses (percents) directly instead of converting to moles first.

4. Percent composition to empirical formula (×2) — a) $K_2Cr_2O_7$

Source: Chapter 8 post, slides 42–43 (step 4: multiply when not whole), slides 35–38
Homework match: HW21 — hw21ex2.png (three-element percent compositions: $MgSO_4$, $SiCl_2Br_2$)
What it tests: The 4-step method: grams → moles → divide by the smallest → whole numbers (multiply every element if needed).

Step 1: percent to grams. Pretend you have exactly **100 g** of the compound, so each percent becomes that many grams: 26.58 g K, 35.35 g Cr, 38.07 g O.

Step 2: grams to moles for each element. Divide by each element's own atomic mass (periodic table):

$$26.58 \cancel{\text{ g K}} \times \underbrace{\frac{1 \text{ mol K}}{39.10 \cancel{\text{ g K}}}}_{\text{atomic mass of K}} = 0.6798 \text{ mol K}$$

$$35.35 \cancel{\text{ g Cr}} \times \underbrace{\frac{1 \text{ mol Cr}}{52.00 \cancel{\text{ g Cr}}}}_{\text{atomic mass of Cr}} = 0.6798 \text{ mol Cr}$$

$$38.07 \cancel{\text{ g O}} \times \underbrace{\frac{1 \text{ mol O}}{16.00 \cancel{\text{ g O}}}}_{\text{atomic mass of O}} = 2.379 \text{ mol O}$$

Step 3: divide every mole value by the smallest one. The smallest mole value is 0.6798 mol (K). Divide **each** element's moles by it:

Element	Moles (step 2)	÷ smallest (0.6798 mol K)	× 2	Subscript
K	0.6798	1.00	2.00	2
Cr	0.6798	1.00	2.00	2
O	2.379	3.50	7.00	7

Step 4: whole numbers? 3.50 is not within 0.1 of a whole number. It ends in a decimal near .5 (halves), so multiply **every** result by 2, including the elements that were already whole (last columns of the table). Empirical formula: $\mathbf{K_2Cr_2O_7}$.

Why the other choices are wrong:

- b) KCrO: Used the masses (percents) directly instead of converting to moles first.
- c) $KCrO_7$: Multiplied only the fractional element by 2. The same multiplier goes on **every** element.
- d) $KCrO_4$: Rounded the fractional result instead of multiplying (only results within 0.1 of a whole number may be rounded).
- e) $K_4Cr_4O_{14}$: Not the **smallest** whole-number ratio; every subscript can still be divided by 2.

5. Grams to empirical formula — d) Na_2SO_3

Source: Chapter 8 post, slides 45–46 (empirical formula from masses), slide 42
Homework match: HW21 — empirical formula from composition data
What it tests: The 4-step method: grams → moles → divide by the smallest → whole numbers (multiply every element if needed).

Step 1: grams of each element. The grams are given: 2.30 g Na, 1.60 g S, 2.40 g O.

Step 2: grams to moles for each element. Divide by each element's own atomic mass (periodic table):

$$2.30 \cancel{\text{ g Na}} \times \underbrace{\frac{1 \text{ mol Na}}{22.99 \cancel{\text{ g Na}}}}_{\text{atomic mass of Na}} = 0.1000 \text{ mol Na}$$

$$1.60 \cancel{\text{ g S}} \times \underbrace{\frac{1 \text{ mol S}}{32.07 \cancel{\text{ g S}}}}_{\text{atomic mass of S}} = 0.04989 \text{ mol S}$$

$$2.40 \cancel{\text{ g O}} \times \underbrace{\frac{1 \text{ mol O}}{16.00 \cancel{\text{ g O}}}}_{\text{atomic mass of O}} = 0.1500 \text{ mol O}$$

Step 3: divide every mole value by the smallest one. The smallest mole value is 0.04989 mol (S). Divide **each** element's moles by it:

Element	Moles (step 2)	÷ smallest (0.04989 mol S)	Subscript
Na	0.1000	2.01	2
S	0.04989	1.00	1
O	0.1500	3.01	3

Step 4: whole numbers? Every result is within 0.1 of a whole number, so round them. No multiplier is needed. Empirical formula: $\mathbf{Na_2SO_3}$.

Why the other choices are wrong:

- a) Na_3SO_2: Subscripts swapped between the elements.
- b) NaSO: Used the masses (percents) directly instead of converting to moles first.
- c) $Na_4S_2O_6$: Not the **smallest** whole-number ratio; every subscript can still be divided by 2.
- e) Na_3SO_3: Miscounted Na: its step-3 result is 2.01, which gives 2, not 3.

6. Empirical to molecular formula — e) C_9H_{12}

Source: Chapter 8 post, slide 48 (molecular formula = $(\text{empirical formula})_n$), slide 49 (LA Practice), slide 47
Homework match: HW21 — hw21ex3.png (empirical formula + molar mass → molecular formula)
What it tests: Molecular formula = $(\text{empirical formula})_n$ with n = molar mass ÷ empirical formula mass.

Step 1: empirical formula mass. Add up the atomic masses in the empirical formula: C_3H_4 = 3(12.01) + 4(1.008) = 40.06 g/mol.

Step 2: find n.

$$n = \frac{\text{molar mass of the compound}}{\text{empirical formula mass}} = \frac{120.19 \text{ g/mol}}{40.06 \text{ g/mol}} = 3.00 \approx 3$$

Step 3: multiply every subscript by n. $(C_3H_4)_3$: C 3 × 3 = 9, H 4 × 3 = 12. Molecular formula: $\mathbf{C_9H_{12}}$.

Why the other choices are wrong:

- a) C_6H_8: n = 2 would give 80.12 g/mol.
- b) $C_{12}H_{16}$: n = 4 would give a molar mass of 160.25 g/mol.
- c) C_9H_4: Multiplied only the first subscript by n = 3. **Every** subscript gets multiplied.
- d) C_3H_4: That is the empirical formula itself (n = 1). Its molar mass would be 40.06 g/mol, not 120.19.

7. From percent composition to a mole ratio — c) 0.600 mol

Source: Chapter 8 post, slides 42–49 (empirical then molecular formula); Chapter 9 post, slide 9 (balancing a combustion), slides 10–11
Homework match: HW21 — hw21ex3.png; HW22 — hw22ex3.png ($CH_3CH_2OH + O_2$ equations)
What it tests: Section 8.7 feeding Chapter 9: percent composition → empirical → molecular formula → balanced equation → mole ratio.

Step 1: percent to grams. Pretend you have exactly **100 g** of the compound, so each percent becomes that many grams: 52.14 g C, 13.13 g H, 34.73 g O.

Step 2: grams to moles for each element. Divide by each element's own atomic mass (periodic table):

$$52.14 \cancel{\text{ g C}} \times \underbrace{\frac{1 \text{ mol C}}{12.01 \cancel{\text{ g C}}}}_{\text{atomic mass of C}} = 4.341 \text{ mol C}$$

$$13.13 \text{ g H} \times \underbrace{\frac{1 \text{ mol H}}{1.008 \text{ g H}}}_{\text{atomic mass of H}} = 13.03 \text{ mol H}$$

$$34.73 \text{ g O} \times \underbrace{\frac{1 \text{ mol O}}{16.00 \text{ g O}}}_{\text{atomic mass of O}} = 2.171 \text{ mol O}$$

Step 3: divide every mole value by the smallest one. The smallest mole value is 2.171 mol (O). Divide **each** element's moles by it:

Element	Moles (step 2)	÷ smallest (2.171 mol O)	Subscript
C	4.341	2.00	2
H	13.03	6.00	6
O	2.171	1.00	1

Step 4: whole numbers? Every result is within 0.1 of a whole number, so round them. No multiplier is needed. Empirical formula: $\mathbf{C_2H_6O}$.

Step 5: molecular formula. Empirical formula mass: $C_2H_6O = 2(12.01) + 6(1.008) + 16.00 = 46.07$ g/mol. n = 46.07 ÷ 46.07 = 1.00 ≈ 1, so multiply every subscript by 1: $\mathbf{C_2H_6O}$.

Step 6: write the balanced equation. Combustion of C_2H_6O: C first ($2CO_2$), then H ($3H_2O$, 6 H). The right side has 2(2) + 3(1) = 7 O. The fuel already supplies **1** O, so O_2 must supply 6 O: $3O_2$. $\mathbf{C_2H_6O + 3O_2 \rightarrow 2CO_2 + 3H_2O}$.

Step 7: moles $\mathbf{C_2H_6O}$ to moles $\mathbf{O_2}$. Have C_2H_6O (coefficient **1**), want O_2 (coefficient **3**).

$$0.2000 \text{ mol } C_2H_6O \times \underbrace{\frac{3 \text{ mol } O_2}{1 \text{ mol } C_2H_6O}}_{\text{coefficient of } O_2 \text{ / coefficient of } C_2H_6O} = \boxed{0.600 \text{ mol } O_2}$$

The molecular formula matters here. A mole of the **empirical** unit is a different amount of substance from a mole of the real molecule, so it would give a different number of moles of product.

Why the other choices are wrong:

- a) 0.200 mol: Assumed 1 : 1. O_2's coefficient is 3.
- b) 0.400 mol: Used CO_2's coefficient (2) instead of O_2's (3).
- d) 0.700 mol: Balanced O without counting the O already in the fuel (3.5 O_2 instead of 3).
- e) 1.20 mol: Assumed n = 2 ($C_4H_{12}O_2$). Here n = 46.07 ÷ 46.07 = 1.

8. What a mole–mole conversion needs — a) The coefficients of A and B in the balanced equation

Source: Chapter 9 post, slides 5–7 (mole ratios from the balanced equation), slide 10 (Mole–Mole Conversions)
Homework match: HW22 — hw22ex2.png (choosing the mole ratio from the coefficients)
What it tests: The mole ratio comes only from the coefficients of a balanced equation.

The **mole ratio** (slides 5–7) is the only conversion factor that connects two different substances, and it comes from the **coefficients** of the balanced equation:

$$\text{mol B} = \text{mol A} \times \frac{\text{coefficient of B}}{\text{coefficient of A}}$$

Why the other choices are wrong:

- b) Avogadro's number: That converts moles particles. It doesn't relate two different substances.
- c) The masses of A and B present: Masses are needed to find moles, but the link from A to B is the mole ratio.
- d) The molar masses of A and B: Molar masses convert grams moles of the **same** substance, not A to B.
- e) The subscripts in the formulas of A and B: Subscripts count atoms within one formula. The ratio between substances comes from the coefficients.

9. Mole–mole conversion — a) 1.20 mol

Source: Chapter 9 post, slides 10–11 (Mole–Mole Conversions Practice)
Homework match: HW22 — hw22ex3.png (moles of product from moles of another substance)
What it tests: mol → mol → mole ratio → mol.

Step 1: moles NH_3 to moles N_2. Have NH_3 (coefficient **4**), want N_2 (coefficient **2**). The question asks for **moles**, so stop here.

$$2.400 \ \cancel{\text{mol NH}_3} \times \underbrace{\frac{2 \text{ mol N}_2}{4 \ \cancel{\text{mol NH}_3}}}_{\text{coefficient of N}_2 \ / \ \text{coefficient of NH}_3} = \boxed{1.20 \text{ mol N}_2}$$

Why the other choices are wrong:

- b) 1.80 mol: Used O_2's coefficient (3) instead of N_2's (2).
- c) 2.40 mol: Assumed 1 : 1. The coefficients are 4 (NH_3) and 2 (N_2).
- d) 4.80 mol: Mole ratio upside down: × 4/2 instead of × 2/4.
- e) 33.6 mol: Multiplied by the molar mass of N_2. That gives grams, not moles.

10. Grams to moles of another substance — c) 0.399 mol

Source: Chapter 9 post, slide 18 (Stoichiometry Practice 2: moles of H_2SO_4 from 19.9 g KOH), slide 14
Homework match: HW22 — hw22ex3.png (moles needed from grams: 0.206 g Al → mol Br_2)
What it tests: g → mol → mole ratio → mol.

Step 1: molar masses. $CS_2 = 12.01 + 2(32.07) = 76.15$ g/mol.

Step 2: grams CS_2 to moles CS_2.

$$15.2 \text{ g } \cancel{CS_2} \times \underbrace{\frac{1 \text{ mol } CS_2}{76.15 \text{ g } \cancel{CS_2}}}_{\text{molar mass of } CS_2} = 0.1996 \text{ mol } CS_2$$

Step 3: moles CS_2 to moles SO_2. Have CS_2 (coefficient **1**), want SO_2 (coefficient **2**). The question asks for **moles**, so stop here.

$$0.1996 \text{ } \cancel{\text{mol } CS_2} \times \underbrace{\frac{2 \text{ mol } SO_2}{1 \text{ } \cancel{\text{mol } CS_2}}}_{\text{coefficient of } SO_2 \text{ / coefficient of } CS_2} = \boxed{0.399 \text{ mol } SO_2}$$

Why the other choices are wrong:

- a) 0.0998 mol: Mole ratio upside down: × 1/2 instead of × 2/1.
- b) 0.200 mol: Assumed 1 : 1. The coefficients are 1 (CS_2) and 2 (SO_2).
- d) 0.599 mol: Used O_2's coefficient (3) instead of SO_2's (2).
- e) 25.6 mol: Multiplied by the molar mass of SO_2. That gives grams, not moles.

11. Grams to grams — e) 17.0 g

Source: Chapter 9 post, slides 19–20 (Stoichiometry Practice: g → mol → mol → g), slide 14
Homework match: HW23 — hw23ex1.png (grams of product from grams of reactant)
What it tests: g → mol → mole ratio → g.

Step 1: molar masses. Cu = 63.55 g/mol (periodic table); Ag = 107.87 g/mol (periodic table).

Step 2: grams Cu to moles Cu.

$$5.00 \text{ g } \cancel{\text{Cu}} \times \underbrace{\frac{1 \text{ mol Cu}}{63.55 \text{ g } \cancel{\text{Cu}}}}_{\text{molar mass of Cu}} = 0.07868 \text{ mol Cu}$$

Step 3: moles Cu to moles Ag. Have Cu (coefficient **1**), want Ag (coefficient **2**).

$$0.07868 \text{ } \cancel{\text{mol Cu}} \times \underbrace{\frac{2 \text{ mol Ag}}{1 \text{ } \cancel{\text{mol Cu}}}}_{\text{coefficient of Ag / coefficient of Cu}} = 0.1574 \text{ mol Ag}$$

Step 4: moles Ag to grams Ag.

$$0.1574 \text{ } \cancel{\text{mol Ag}} \times \underbrace{\frac{107.87 \text{ g Ag}}{1 \text{ } \cancel{\text{mol Ag}}}}_{\text{molar mass of Ag}} = \boxed{17.0 \text{ g Ag}}$$

Why the other choices are wrong:

- a) 0.157 g: That is the **moles** of Ag. The last step (× molar mass) is missing.
- b) 4.24 g: Mole ratio upside down: × 1/2 instead of × 2/1.
- c) 8.49 g: Assumed 1 : 1. The coefficients are 1 (Cu) and 2 (Ag).
- d) 10.0 g: Used the molar mass of Cu in the last step instead of Ag's.

12. Balance first, then grams to grams — d) 9.42 g

Source: Chapter 9 post, slide 16 (balance before using coefficients), slide 14, slide 19
Homework match: HW22 — hw22ex1.png (balance with the lowest whole-number coefficients); HW23 — hw23ex3.png
What it tests: g → mol → mole ratio → g, after balancing.

Step 1: balance the equation. K_2O has 1 O, and O_2 supplies 2 O, so make **2**K_2O. That has 4 K: **4K + O_2 → 2K_2O**.

Step 2: molar masses. K = 39.10 g/mol (periodic table); K_2O = 2(39.10) + 16.00 = 94.20 g/mol.

Step 3: grams K to moles K.

$$7.82 \cancel{\text{ g K}} \times \underbrace{\frac{1 \text{ mol K}}{39.10 \cancel{\text{ g K}}}}_{\text{molar mass of K}} = 0.2000 \text{ mol K}$$

Step 4: moles K to moles K_2O. Have K (coefficient **4**), want K_2O (coefficient **2**).

$$0.2000 \cancel{\text{ mol K}} \times \underbrace{\frac{2 \text{ mol } K_2O}{4 \cancel{\text{ mol K}}}}_{\text{coefficient of } K_2O \text{ / coefficient of K}} = 0.1000 \text{ mol } K_2O$$

Step 5: moles K_2O to grams K_2O.

$$0.1000 \cancel{\text{ mol } K_2O} \times \underbrace{\frac{94.20 \text{ g } K_2O}{1 \cancel{\text{ mol } K_2O}}}_{\text{molar mass of } K_2O} = \boxed{9.42 \text{ g } K_2O}$$

Why the other choices are wrong:

- a) 0.100 g: That is the **moles** of K_2O. The last step (× molar mass) is missing.
- b) 3.91 g: Used the molar mass of K in the last step instead of K_2O's.
- c) 4.71 g: Used O_2's coefficient (1) instead of K_2O's (2).
- e) 18.8 g: Assumed 1 : 1. The coefficients are 4 (K) and 2 (K_2O).

13. Limiting reactant and leftover — b) O_2 is limiting, and 9.40 g of ZnS is left over

Source: Chapter 9 post, slide 24 (limiting reactant from two masses), slides 21–23
Homework match: HW23 — hw23ex2.png (5.65 g of each reactant: which is limiting?)
What it tests: Limiting reactant by need-vs-have in moles, then the leftover mass of the excess reactant.

Step 1: molar masses. ZnS = 65.38 + 32.07 = 97.45 g/mol; O_2 = 2(16.00) = 32.00 g/mol.

Step 2: grams to moles for each reactant.

$$50.0 \text{ g } \cancel{\text{ZnS}} \times \underbrace{\frac{1 \text{ mol ZnS}}{97.45 \text{ g } \cancel{\text{ZnS}}}}_{\text{molar mass of ZnS}} = 0.5131 \text{ mol ZnS}$$

$$20.0 \text{ g } \cancel{\text{O}_2} \times \underbrace{\frac{1 \text{ mol } \text{O}_2}{32.00 \text{ g } \cancel{\text{O}_2}}}_{\text{molar mass of } \text{O}_2} = 0.6250 \text{ mol } \text{O}_2$$

Step 3: how much O_2 does all the ZnS need? Have ZnS (coefficient **2**), want O_2 (coefficient **3**).

$$0.5131 \cancel{\text{ mol ZnS}} \times \underbrace{\frac{3 \text{ mol } \text{O}_2}{2 \cancel{\text{ mol ZnS}}}}_{\text{coefficient of } \text{O}_2 \text{ / coefficient of ZnS}} = 0.7696 \text{ mol } \text{O}_2$$

Step 4: compare. You **need** 0.7696 mol O_2 but only **have** 0.6250 mol, so **O_2 is limiting** and ZnS is in excess.

Step 5: how much ZnS reacts? All 0.6250 mol of the limiting O_2 reacts. Have O_2 (coefficient **3**), want ZnS (coefficient **2**).

$$0.6250 \cancel{\text{ mol O}_2} \times \underbrace{\frac{2 \text{ mol ZnS}}{3 \cancel{\text{ mol O}_2}}}_{\text{coefficient of ZnS / coefficient of } \text{O}_2} = 0.4167 \text{ mol ZnS}$$

$$0.4167 \cancel{\text{ mol ZnS}} \times \underbrace{\frac{97.45 \text{ g ZnS}}{1 \cancel{\text{ mol ZnS}}}}_{\text{molar mass of ZnS}} = 40.6 \text{ g ZnS}$$

Step 6: left over. 50.0 g ZnS at the start − 40.6 g that reacts = **9.40 g left over**.

Mass alone never decides the limiting reactant: compare in moles, through the mole ratio.

Why the other choices are wrong:

- a) ZnS is limiting, and O_2 is left over: Check what all of one reactant needs. The need-vs-have comparison shows the O_2 runs out first.
- c) O_2 is limiting, and 50.0 g of ZnS is left over: That is **all** of the ZnS. Some of it reacts with the O_2, so less is left.
- d) O_2 is limiting, and 40.6 g of ZnS is left over: That is the mass of ZnS that **reacts**. Left over = 50.0 g − 40.6 g.
- e) Neither is limiting; both reactants are used up completely: That needs the moles to be in exactly a 2 : 3 ratio, and they aren't.

14. Theoretical yield from two masses — b) 15.9 g

Source: Chapter 9 post, slide 27 (theoretical yield when both masses are given), slide 25, slide 24
Homework match: HW23 — hw23ex3.png (limiting reactant, then mass of product); HW24 — hw24ex2.png
What it tests: Moles of each reactant → limiting reactant → product from the limiting reactant only.

Step 1: molar masses. Al = 26.98 g/mol (periodic table); $CuSO_4$ = 63.55 + 32.07 + 4(16.00) = 159.62 g/mol; Cu = 63.55 g/mol (periodic table).

Step 2: grams to moles for each reactant.

$$5.40 \text{ g Al} \times \underbrace{\frac{1 \text{ mol Al}}{26.98 \text{ g Al}}}_{\text{molar mass of Al}} = 0.2001 \text{ mol Al}$$

$$40.0 \text{ g } CuSO_4 \times \underbrace{\frac{1 \text{ mol } CuSO_4}{159.62 \text{ g } CuSO_4}}_{\text{molar mass of } CuSO_4} = 0.2506 \text{ mol } CuSO_4$$

Step 3: how much $CuSO_4$ does all the Al need? Have Al (coefficient **2**), want $CuSO_4$ (coefficient **3**).

$$0.2001 \text{ mol Al} \times \underbrace{\frac{3 \text{ mol } CuSO_4}{2 \text{ mol Al}}}_{\text{coefficient of } CuSO_4 \text{ / coefficient of Al}} = 0.3002 \text{ mol } CuSO_4 \text{ needed}$$

Step 4: compare. You **need** 0.3002 mol $CuSO_4$ but only **have** 0.2506 mol. The $CuSO_4$ runs out first, so **$CuSO_4$ is limiting** and Al is in excess.

Step 5: theoretical yield from the limiting reactant ($CuSO_4$). Have $CuSO_4$ (coefficient **3**), want Cu (coefficient **3**).

$$0.2506 \text{ mol } CuSO_4 \times \underbrace{\frac{3 \text{ mol Cu}}{3 \text{ mol } CuSO_4}}_{\text{coefficient of Cu / coefficient of } CuSO_4} = 0.2506 \text{ mol Cu}$$

$$0.2506 \text{ mol Cu} \times \underbrace{\frac{63.55 \text{ g Cu}}{1 \text{ mol Cu}}}_{\text{molar mass of Cu}} = \boxed{15.9 \text{ g Cu}}$$

Cross-check: all the Al would make 19.1 g Cu. The smaller amount, 15.9 g, is the theoretical yield.

Why the other choices are wrong:

- a) 0.251 g: That is the **moles** of Cu.
- c) 19.1 g: Calculated from Al. Al is in excess, so this is more than the $CuSO_4$ can make.
- d) 40.0 g: Used the molar mass of $CuSO_4$ instead of Cu in the last step.
- e) 45.4 g: Added the reactant masses. Part of the excess reactant is left over.

15. Percent yield — c) 88.1%

Source: Chapter 9 post, slide 30 (percent yield formula), slides 31–32, slide 28
Homework match: HW24 — hw24ex1.png (percent yield of magnesium oxide)
What it tests: Two parts: theoretical yield from the limiting reactant, then actual / theoretical × 100.

Step 1: which reactant controls the yield? O_2 is in excess, so **Mg is limiting** and the theoretical yield is calculated from it.

Step 2: molar masses. Mg = 24.31 g/mol (periodic table); MgO = 24.31 + 16.00 = 40.31 g/mol.

Step 3: theoretical yield of MgO. Have Mg (coefficient **2**), want MgO (coefficient **2**).

$$4.86 \text{ g Mg} \times \underbrace{\frac{1 \text{ mol Mg}}{24.31 \text{ g Mg}}}_{\text{molar mass of Mg}} = 0.1999 \text{ mol Mg}$$

$$0.1999 \text{ mol Mg} \times \underbrace{\frac{2 \text{ mol MgO}}{2 \text{ mol Mg}}}_{\text{coefficient of MgO / coefficient of Mg}} = 0.1999 \text{ mol MgO}$$

$$0.1999 \text{ mol MgO} \times \underbrace{\frac{40.31 \text{ g MgO}}{1 \text{ mol MgO}}}_{\text{molar mass of MgO}} = 8.059 \text{ g MgO (theoretical)}$$

Step 4: percent yield. The actual yield (7.1 g, measured) goes on top, the theoretical yield on the bottom:

$$\text{percent yield} = \frac{7.1 \text{ g (actual)}}{8.059 \text{ g (theoretical)}} \times 100\% = \boxed{88.1\%}$$

Why the other choices are wrong:

- a) 0.881%: Forgot to multiply by 100.
- b) 17.6%: Divided the actual mass by the molar mass of MgO (40.31) instead of by the theoretical yield.
- d) 114%: Upside down: theoretical / actual. A yield can't be over 100%.
- e) 146%: Divided by the mass of Mg. The bottom must be the theoretical yield of MgO.